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W9023

CMOS LSI

Capacitive Touch and Pen Control Microcontroller

Product Specification

Ver.1.0

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2. Revision History

Doc.Version	Revision Description	Date
1.0	1st release	2021.02.10

3. General Description

- W9023 is the successor product of W9017, as a multi-touch and Wacom ActiveES pen control LSI. LSI architecture is big changed, e.g. CPU performance, Touch-control, AES Pen-control and the other function.

- Major new features of W9023 are
  1) Improved performance of CPU and peripheral functions
     twice CPU performance, bigger flash memory size, flexible quad SPI i/f, Powerful inside step up DCDC converter
  2) Improved of ActiveES pen performance.
     Three receiving circuits run in parallel,
     Dynamic signal processing independent of input signal amplitude,
     Stabilized noise masking function
  3) Improved of touch performance.
     In terms of specifications, the same performance as previous devices are maintained, and detailed characteristics are improved for touch control stability up.

(5.0mm×7.2mm)

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4. Features

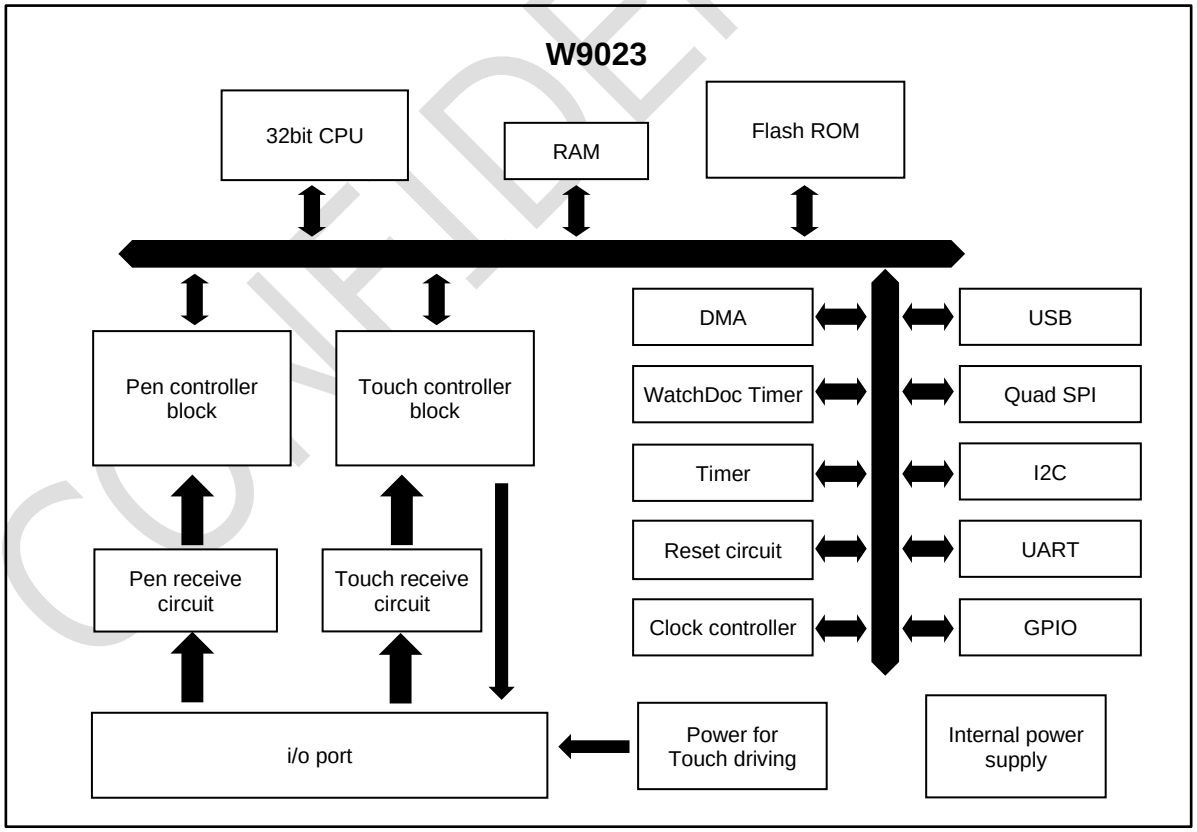
- Operating voltage range **:** VDD=3.0V to 3.6V
- Digital power supply / Analog power supply
  - internal power on reset circuite
  - internal voltage detection circuit

- CPU and peripheral
  - CPU 96MHz operation
  - USB2.0 Full-Speed device controller
  - I2C Fast mode (400kHz), Fast mode+ (1MHz)
  - Quad SPI (Slave, up to 30MHz)
  - Touch controller
  - ActiveES pen controller
  - High voltage (up to 9V) driving signals for a touch sensor

- Operating Temperature Range: -30°C to +85°C

5. Block Diagram

Block diagram is shown below.



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6. Pinout

【Top View】

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	RX25	RX27	RX29	RX31	RX33	RX35	RX39	RX45	RX51	RX57	RX63	RX69	VREFL 2	VREFL 1	AVSS _RX1	AVDD _RX1
B	RX24	RX26	RX28	RX30	RX32	RX34	RX38	RX44	RX50	RX56	RX62	RX68	VREFH1	V12 _AFE	V12 _IN	V12 _OUT
C	RX23	RX22	RX21	RX20	RX19	RX18	RX37	RX43	RX49	RX55	RX61	RX67	VREFH2	DVSS _VOL	TEST3	TEST4
D	RX12	RX13	RX14	RX15	RX16	RX17	RX36	RX42	RX48	RX54	RX60	RX66	PB6	PB7	DVSS _USB	DVDD _USB
E	AVSS _RX2	RX11	RX10	RX09	RX08	RX07	RX06	RX41	RX47	RX53	RX59	RX65	RX71	PB0	PB4 (SDA)	DP
F	VREF16	RX00	RX01	RX02	RX03	RX04	RX05	RX40	RX46	RX52	RX58	RX64	RX70	N.C	PB5 (SCL)	DM
G	AVDD _RX2	TEST0	TEST1	TEST2	MODE	RESET _N	PC2	PC1	PC0	PB3	PB2	PB1	PA3	N.C	PA0	DVDD _SPI
H	TX00	TX07	TX08	TX15	TX16	TX23	TX24	TX30	TX31	TX36	TX37	PA7	PC3	PA4	PA1	X2
J	TX01	TX06	TX09	TX14	TX17	TX22	TX25	TX29	TX32	TX35	TX38	TX43	PA2	N.C	PA6	X1
K	TX02	TX05	TX10	TX13	TX18	TX21	TX26	TX28	TX33	TX34	TX39	TX42	TX44	ISNSP _IN	PA5	DVSS
L	TX03	TX04	TX11	TX12	TX19	TX20	TX27	AVDD _TX	AVSS _TX	VTX	TX40	TX41	TX45	PVSS	DCSW	DVDD 33

- Red font : Power supply terminal
- Blue font : Ground terminal
- Green font : Power-related terminal

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7. Pin Description

Function	Pin number	Name	Type	Description
Port A	G15	PA0	I/O	General purpose I/O port
	H15	PA1	I/O	
	J13	PA2	I/O	
	G13	PA3	I/O	
	H14	PA4	I/O	
	K15	PA5	I/O	
	J15	PA6	I/O	
	H12	PA7	I/O	
Port B	E14	PB0	I/O	
	G12	PB1	I/O	
	G11	PB2	I/O	
	G10	PB3	I/O	
	E15	PB4	I/O	
	F15	PB5	I/O	
	D13	PB6	I/O	
	D14	PB7	I/O	
Port C	G9	PC0	I/O	
	G8	PC1	I/O	
	G7	PC2	I/O	
	H13	PC3	I/O	
USB	E16	DP	I/O	USB interface
	F16	DM	I/O	
Clock	J16	X1	I	Crystal oscillator connection
	H16	X2	O	
Reset	G6	RESET_N	I	Reset input (Low active, Pull-up resister)
Power supply	L16	DVDD33	Power	Power supply
	D16	DVDD_USB	Power	Power supply
	G16	DVDD_SPI	Power	Power supply
	L8	AVDD_TX	Power	Power supply for TX drive
	A16	AVDD_RX1	Power	power supply for Analog circuit
	G1	AVDD_RX2	Power	power supply for Analog circuit
	L10	VTX	Power	Power supply for internal DCDC
	L15	DCSW	Power	Connect to PVSS
	K14	ISNSP_IN	Power	External FET connection for internal DCDC
	F1	VREF16	Power	Analog reference voltage
	B16	V12_OUT	Power	1.2V output
	B15	V12_IN	Power	1.2V input for digital kernel
	B14	V12_AFE	Power	1.2V input for Analog circuit
	B13	VREFH1	Power	1.2V input for Analog circuit
	C13	VREFH2	Power	1.2V input for Analog circuit

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Function	Pin number	Name	Type	Description
GND	K16	DVSS	GND	Digital ground
	D15	DVSS_USB	GND	Digital ground
	A15	AVSS_RX1	GND	Analog ground
	E1	AVSS_RX2	GND	Analog ground
	L9	AVSS_TX	GND	Analog ground
	L14	PVSS	GND	Ground for internal DC/DC
	C14	DVSS_VOL	GND	Ground for internal LDO
	A14	VREFL1	GND	Analog ground
	A13	VREFL2	GND	Analog ground
TX	H1	TX00	I/O	Touch operation : Driving signal output Pen operation : Receive signal input
	J1	TX01	I/O	
	K1	TX02	I/O	
	L1	TX03	I/O	
	L2	TX04	I/O	
	K2	TX05	I/O	
	J2	TX06	I/O	
	H2	TX07	I/O	
	H3	TX08	I/O	
	J3	TX09	I/O	
	K3	TX10	I/O	
	L3	TX11	I/O	
	L4	TX12	I/O	
	K4	TX13	I/O	
	J4	TX14	I/O	
	H4	TX15	I/O	
	H5	TX16	I/O	
	J5	TX17	I/O	
	K5	TX18	I/O	
	L5	TX19	I/O	
	L6	TX20	I/O	
	K6	TX21	I/O	
	J6	TX22	I/O	
	H6	TX23	I/O	
	H7	TX24	I/O	
	J7	TX25	I/O	
	K7	TX26	I/O	
	L7	TX27	I/O	
	K8	TX28	I/O	
	J8	TX29	I/O	
	H8	TX30	I/O	
	H9	TX31	I/O	
	J9	TX32	I/O	
	K9	TX33	I/O	
	K10	TX34	I/O	
	J10	TX35	I/O	

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Function	Pin number	Name	Type	Description
TX	H10	TX36	I/O	Touch operation : Driving signal output Pen operation : Receive signal input
	H11	TX37	I/O	
	J11	TX38	I/O	
	K11	TX39	I/O	
	L11	TX40	I/O	
	L12	TX41	I/O	
	K12	TX42	I/O	
	J12	TX43	I/O	
	K13	TX44	I/O	
	L13	TX45	I/O	
RX	F2	RX00	I/O	Touch operation : Receive signal input Pen operation : Receive signal input
	F3	RX01	I/O	
	F4	RX02	I/O	
	F5	RX03	I/O	
	F6	RX04	I/O	
	F7	RX05	I/O	
	E7	RX06	I/O	
	E6	RX07	I/O	
	E5	RX08	I/O	
	E4	RX09	I/O	
	E3	RX10	I/O	
	E2	RX11	I/O	
	D1	RX12	I/O	
	D2	RX13	I/O	
	D3	RX14	I/O	
	D4	RX15	I/O	
	D5	RX16	I/O	
	D6	RX17	I/O	
	C6	RX18	I/O	
	C5	RX19	I/O	
	C4	RX20	I/O	
	C3	RX21	I/O	
	C2	RX22	I/O	
	C1	RX23	I/O	
	B1	RX24	I/O	
	A1	RX25	I/O	
	B2	RX26	I/O	
	A1	RX27	I/O	
	B3	RX28	I/O	
	A3	RX29	I/O	
	B4	RX30	I/O	
	A4	RX31	I/O	
	B5	RX32	I/O	
	A5	RX33	I/O	

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Function	Pin number	Name	Type	Description
RX	B6	RX34	I/O	Touch operation : Receive signal input Pen operation : Receive signal input
	A6	RX35	I/O	
	D7	RX36	I/O	
	C7	RX37	I/O	
	B7	RX38	I/O	
	A7	RX39	I/O	
	F8	RX40	I/O	
	E8	RX41	I/O	
	D8	RX42	I/O	
	C8	RX43	I/O	
	B8	RX44	I/O	
	A8	RX45	I/O	
	F9	RX46	I/O	
	E9	RX47	I/O	
	D9	RX48	I/O	
	C9	RX49	I/O	
	B9	RX50	I/O	
	A9	RX51	I/O	
	F10	RX52	I/O	
	E10	RX53	I/O	
	D10	RX54	I/O	
	C10	RX55	I/O	
	B10	RX56	I/O	
	A10	RX57	I/O	
	F11	RX58	I/O	
	E11	RX59	I/O	
	D11	RX60	I/O	
	C11	RX61	I/O	
	B11	RX62	I/O	
	A11	RX63	I/O	
	F12	RX64	I/O	
	E12	RX65	I/O	
	D12	RX66	I/O	
	C12	RX67	I/O	
	B12	RX68	I/O	
	A12	RX69	I/O	
	F13	RX70	I/O	
	E13	RX71	I/O	
Test	G2	TEST0	I	Testing pin (GND connection)
	G3	TEST1	I	Testing pin (GND connection)
	G4	TEST2	I	Testing pin (GND connection)
	G5	MODE	I	Testing pin (GND connection)
	C15	TEST3	---	Testing pin (Not connect)
	C16	TEST4	---	Testing pin (Not connect)

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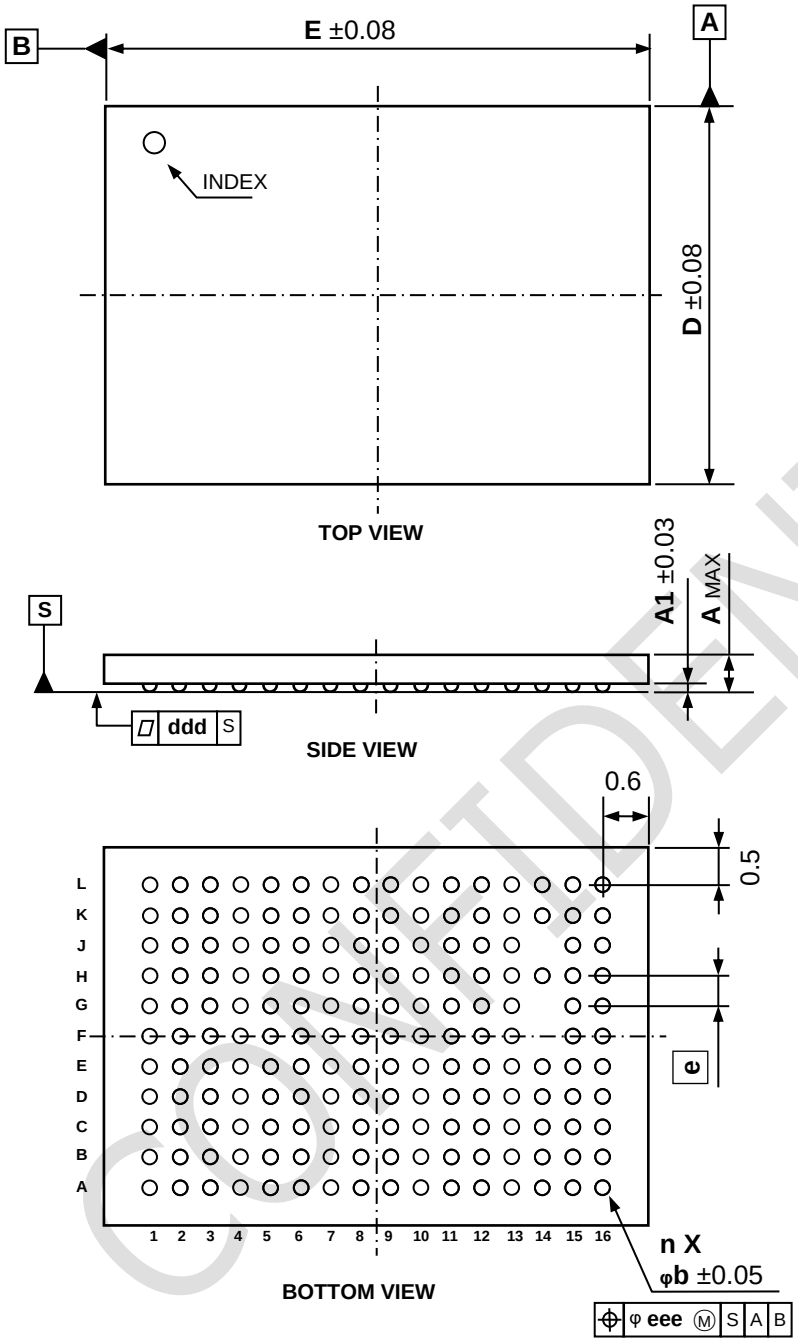
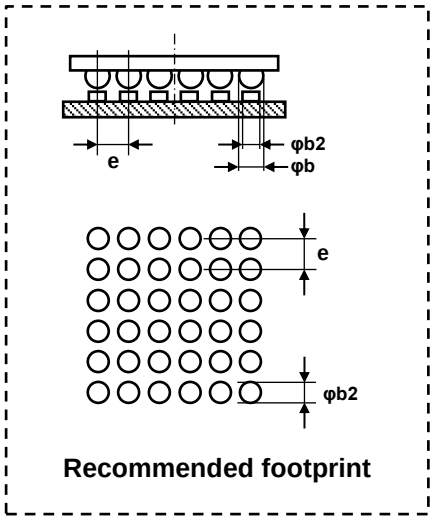
8. Package Outline



unit : mm

Item	Symbol	Dimension [mm]
Total Thickness (MAX)	**A**	0.65
Stand Off	**A1**	0.13
Body Size	**D**	5.0
	E	7.2
Ball Width	**b**	0.22
Ball Pitch	**e**	0.4
Ball Count	**n**	173
Coplanarity	**ddd**	0.08
Ball Offset (Package)	**eee**	0.05
PCB Ball Pad Size	**b2**	0.22



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9. Absolute Maximum Ratings

Ta=25℃, DVSS=DVSS_USB=AVSS_RX1=AVSS_RX2=AVSS_TX=0V

Parameter	Symbol	Applicable Pins	Condition	Rating	Unit
Maximum supply voltage	---	DVDD33, DVDD_USB, DVDD_SPI, AVDD_RX1, AVDD_RX2, AVDD_TX		-0.3 ~ +3.9	V
Input Voltage	V_{IN1}	PA0 ~ PA7, PB0 ~ PB7	input tolerant	-0.3 ~ +3.9	V
	V_{IN2}	PC0 ~ PC3, RESET_N		-0.3 ~ DVDD33+0.3	V
	V_{IN3}	RX00 ~ RX71		-0.3 ~ ADVDD_RX+0.3	V
Pin voltage	---	VTX		-0.3 ~ +9.4	V
Allowable power consumption	PD_{max}		Ta=+85℃	600	mW
Operating temperature	T_{OPR}			-30 ～ +85	℃
Storage temperature	T_{STG}			-50 ～ +125	℃

Absolute maximum ratings are limiting values of operating and environmental conditions.
Stresses above the absolute maximum ratings table may cause damage to the device.
Exposure to conditions beyond those listed above may affect device reliability or cause permanent damage to the device.
The designer should design so that no Absolute maximum rating value is exceeded with respect to current, voltage, power consumption, temperature, etc.

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10. Electrical Characterristics

DC Electrical Characteristics

Ta = -30~+85℃, DVSS=DVSS_VOL=DVSS_USB=AVSS_RX1=AVSS_RX2= AVSS_TX=0V

Parameter	Symbol	Pins	Condition		Rated Value			unit
					Min.	Typ.	Max.	
Operating supply voltage	VDD1	DVDD33, DVDD_USB, AVDD_RX1, AVDD_RX2, AVDD_TX	DVDD33 = DVDD_USB = AVDD_RX1 = AVDD_RX2 = AVDD_TX		3.0	---	3.6	V
	VDD2	DVDD_SPI	1.8V input		1.71	---	1.89	
			3.3V input	at HSPI	3.14	---	3.46	
				at Port	3.0	---	3.6	
Low-level input voltage	V_{IL1}	PA0 ~ PA7, PB0 ~ PB7, RESET_N			DVSS	---	0.4	V
	V_{IL2}	PC0 ~ PC3			DVSS	---	0.25×VDD1	
	V_{IL3}	RX00 ~ RX71			AVSS	---	0.25×VDD1	
	V_{IL4}	DP, DM			DVSS_USB	---	0.8	
High level Input voltage	V_{IH1}	PA0 ~ PA7			0.9	---	VDD2	V
	V_{IH2}	PB0 ~ PB7			0.9	---	VDD1	
	V_{IH3}	PC0 ~ PC3, RESET_N			0.75×VDD1	---	VDD1	
	V_{IH4}	RX00 ~ RX71			0.75×VDD1	---	VDD1	
	V_{IH5}	DP, DM			2.0	---	VDD1	
Low-level output voltage	V_{OL1}	PB0 ~ PB2, PB6, PB7, PC0 ~ PC2	DVDD33=3.0V, I_{OL}=2mA		---	---	0.4	V
	V_{OL2}	PB3	DVDD33=3.0V, I_{OL}=4mA		---	---	0.4	
	V_{OL3}	PB4, PB5	DVDD33=3.0V, I_{OL}=20mA		---	---	0.4	
	V_{OL4}	PC3	DVDD33=3.0V, I_{OL}=8mA		---	---	0.4	
	V_{OL5}	PA0 ~ PA7	DVDD_SPI=1.17V, I_{OL}=3mA		---	---	0.4	
	V_{OL6}		DVDD_SPI=3.0V, I_{OL}=3mA		---	---	0.4	
	V_{OL7}	RX00 ~ RX71	AVDD_RX=3.0V, I_{OL}=4mA		---	---	0.4	
	V_{OL8}	DP, DM			---	---	0.3	
High-level output voltage	V_{OH1}	PB0 ~ PB2, PB4 ~ PB7, PC0 ~ PC2	DVDD33=3.0V, I_{OH}= -2mA		VDD1-0.4	---	---	V
	V_{OH2}	PB3	DVDD33=3.0V, I_{OH}= -4mA		VDD1-0.4	---	---	
	V_{OH3}	PC3	DVDD33=3.0V, I_{OH}= -8mA		VDD1-0.4	---	---	
	V_{OH4}	PA0 ~ PA7	DVDD_SPI=1.17V, I_{OH}= -3mA		VDD2-0.4	---	---	
	V_{OH5}		DVDD_SPI=3.0V, I_{OH}= -3mA			---	---	
	V_{OH6}	RX00 ~ RX71	AVDD_RX=3.0V, I_{OH}= -4mA		VDD1-0.4	---	---	
	V_{OH7}	DP, DM			2.8	---	---	
Pull-up resistor	R_{PU1}	RESET_N	Pull up		0.9	1.35	1.57	kΩ
	R_{PU2}	DP	Pull up		12	23	47	
	R_{PU3}	PA0 ~ PA7, PB0 ~ PB7, RX00 ~ RX71	Pull up		25	45	125	
			Pull down		25	45	125	
Pin capacity	C_P	All pins			---	---	10	pF

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